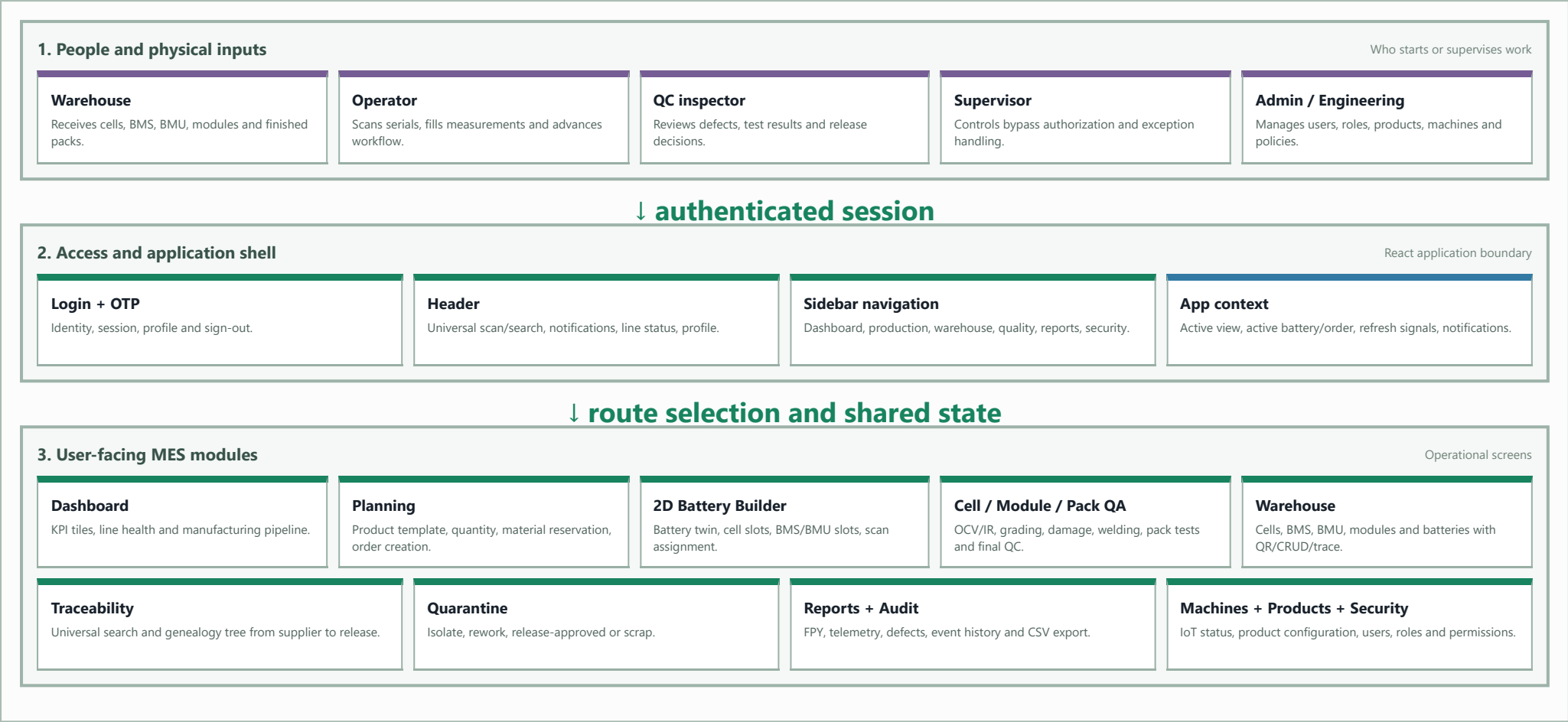


# Power2Go Battery MES: System Block Diagram

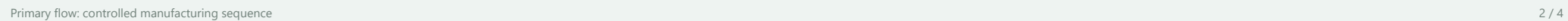
End-to-end architecture and control flow for battery manufacturing, warehouse, quality, traceability, and release.



UI Service/state Data External actor/input

Power2Go Battery MES block diagram

The main manufacturing path is a controlled sequence. Each stage produces records used by the next stage.



# Inventory, QR Passport, and Traceability

The warehouse and genealogy views connect physical serial numbers to a released battery record.

## Warehouse inventory domain

Separate stock views, shared status model

### Cell inventory

Supplier barcode, internal serial, measurements, grade, pallet/box, allocation.

### BMS inventory

Serial, model, supplier, firmware, hardware, protocol, test and status.

### BMU inventory

Serial, model, manufacturer, protocol, test and status.

### Module inventory

Serial, cells, matching score, welding, QC, parent battery.

### Battery inventory

Pack serial, product, modules, BMS, BMU, step, progress, release.

↓ scan / QR / universal identifier

## Battery QR passport

Generated when final QC passes

### Battery label

Pack serial, product, release date/status, stable payload.

### BMS label

BMS serial, model, protocol, parent battery context.

### BMU label

BMU serial, model, protocol, parent battery context.

### Module labels

Module serial, module position, parent battery context.

↓ identifier lookup

## Universal genealogy tree

Bidirectional traceability

### Forward trace

Supplier batch → cell → module → battery → BMS/BMU → final QC → release → dispatch.

### Reverse trace

Scan cell, module, BMS or BMU to find its parent battery, tests, operators and audit events.

### Detail panel

Serial, status, measurement values, timestamps, machine identity, disposition and notes.

## Quarantine and disposition

Safety barrier

### Detect defect

Manual or automated failure creates open quarantine.

### Isolate

Set component status QUARANTINED and prevent allocation.

### Disposition

Release approved, rework, or scrap with signed notes.

### Audit

Record actor, reason, old/new state, time, and resolution.

**2 Integrity rule:** BMS and BMU must link symmetrically to both their own inventory records and the battery record. Quarantine must support both controller types.

# Services, Data, Security, and Audit Boundaries

The lower layers are responsible for consistency, permissions, persistence, integrations, and evidence.

